

LOW VOLTAGE PROTECTION STUDY

1. Nominal Installation Data

Parameter Description	Nominal Values
Installation Type	Trifásica 220/127 V
Demand / Nominal Current (220V)	300 kVA / 787.30A
Transformer Impedance (Z%)	5 %
Environment / Temperature	AMBIENTE / 35 °C

2. Infrastructure and Correction Factors

Installation Parameter	Result
Reference Method	B1
No. of Grouped Circuits	1
Temperature Factor (F1)	0.94
Grouping Factor (F2)	1.0
Total Factor Applied	0.94
<i>Installation: camada única sobre parede, piso, ou em bandeja não perfurada ou prateleira</i>	

3. Sizing of Conductors

Technical Specification	Result
Conductor / Insulation	COBRE / PVC
Phase / Protection Size	1000 mm ² / 500 mm ²
Cores per Pole (Phase/Neutral/Earth)	1
Nominal / Corrected Capacity	906 A / 851.64 A
Cable Length / Voltage Drop	20 m / 0.43 %

4. Protection and Short-Circuit

Fault Analysis	Calculation Memory
Icc3 at Transformer Terminal	15.75 kA
Icc3 at Low Voltage Main Switchboard (LV-MSB)	15.42 kA
Joule Thermal Damage (Time / K Factor)	4.81 s / 103
Circuit Breaker (CB)	787.30 A < CB < 851.64 A